

Sensor Fusion Architecture

Our control logic merges sensor inputs dynamically based on environmental context and priority level:

Sensor	Primary Role	Sensor Fusion & Priority Hierarchy
Camera	Primary Navigation	Calculates relative target angles for cornering, pillar detection, and track color lines. Holds primary control during open track driving.
Rotary Encoders	Odometry & Distance	Tracks precise physical movement to handle accurate parking maneuvers and determine final lap completion.
Time-of-Flight (ToF)	Spatial Safety Net	Continuously monitors close proximity. Serves as a fail-safe hardware override that forces immediate corrective steering if the vehicle approaches a wall, overriding camera inputs.